EX.NO-3.1- CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

AIM:

To create a 3D model from the given orthographic projection using a fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 2GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. First open your Autodesk fusion 360.

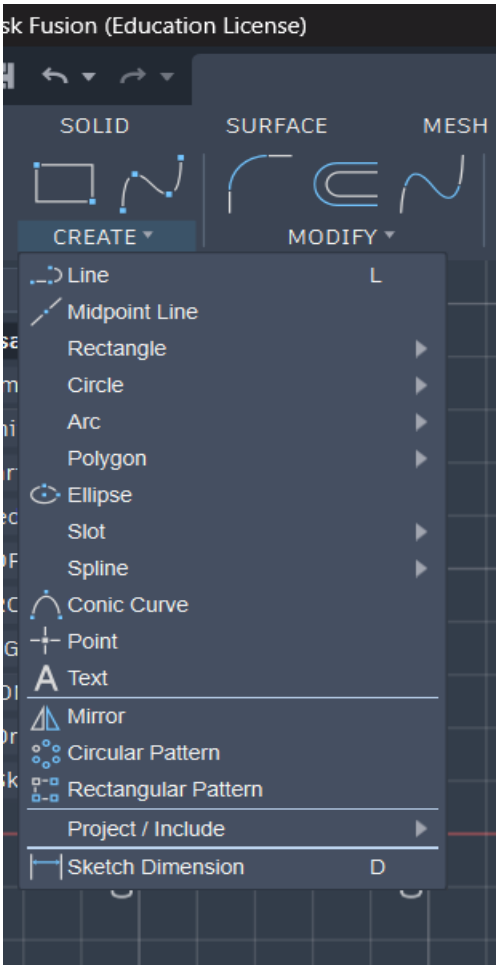
2. Select the new part design.

3. Open a new file and select the sketch icon.

4. Select the plane for sketching in top view.

5. Create several lines making it look like a box.

6. Draw a vertical line of 11mm and a horizontal line of 52mm. Then, draw a line downwards of 40mm.



7. Do the same as above at the bottom and draw a horizontal line of 41mm on both parts and join them.

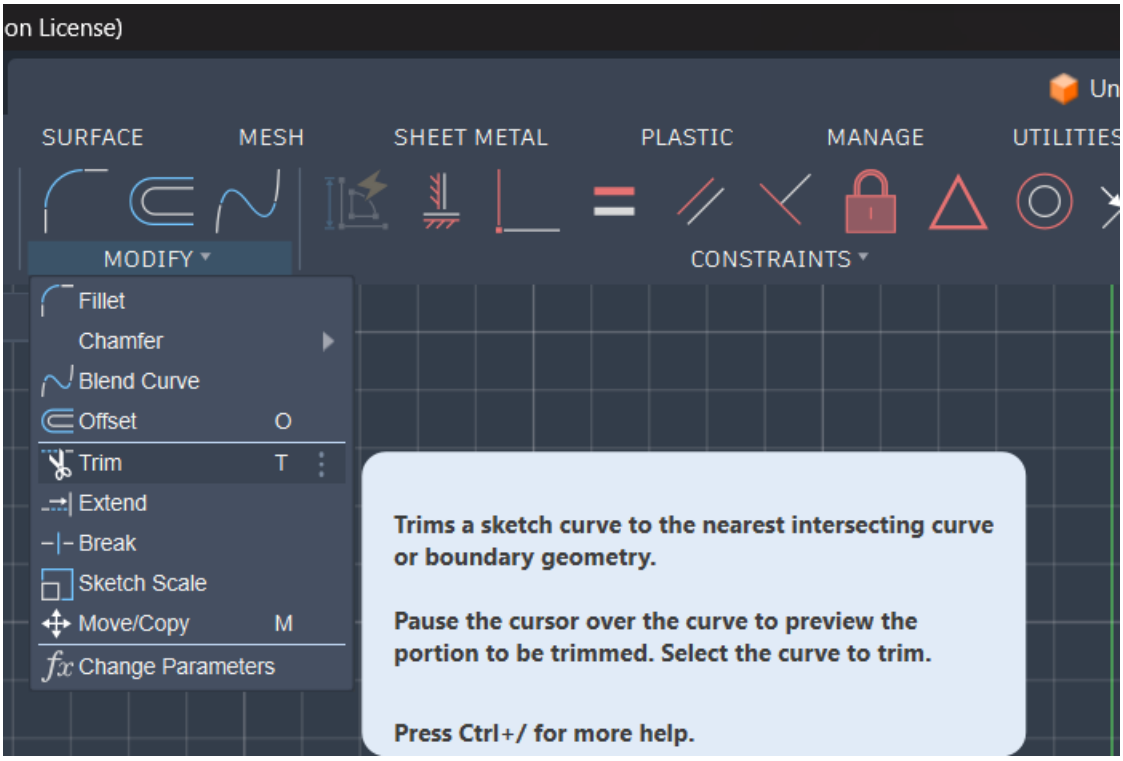
8. Extrude this whole part with a thickness of 32mm.

9. Now, select the right view for sketching.

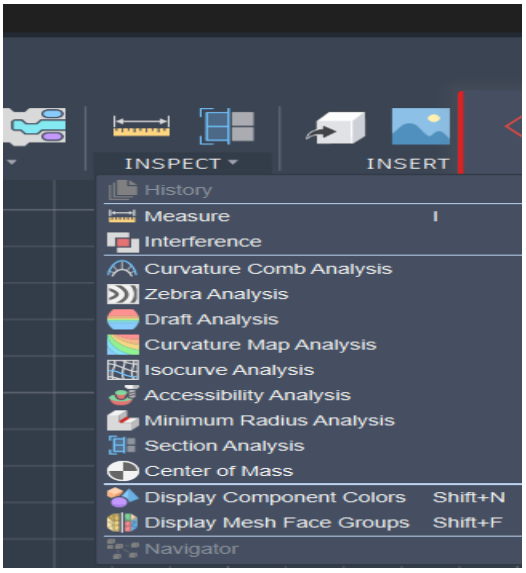
10. Draw a circle in the center with a dimension of 18mm.

11. Extrude this circle with a thickness of 32mm.

12. Then, select the front view for sketching.



13. Draw a circle on the side with a dimension of 16mm and the center is 36mm from the border of the rectangle.

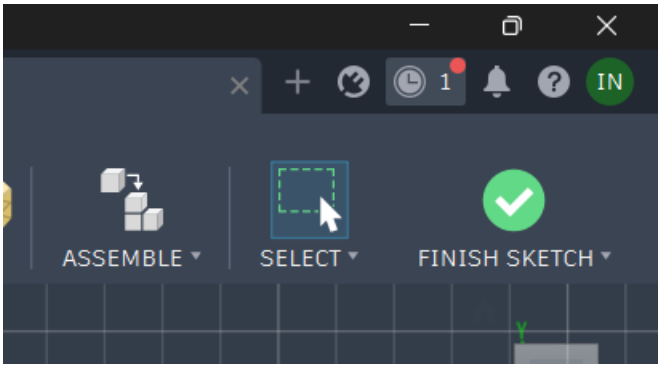


14. When extruding, choose the cut operation to create a hole on both sides.

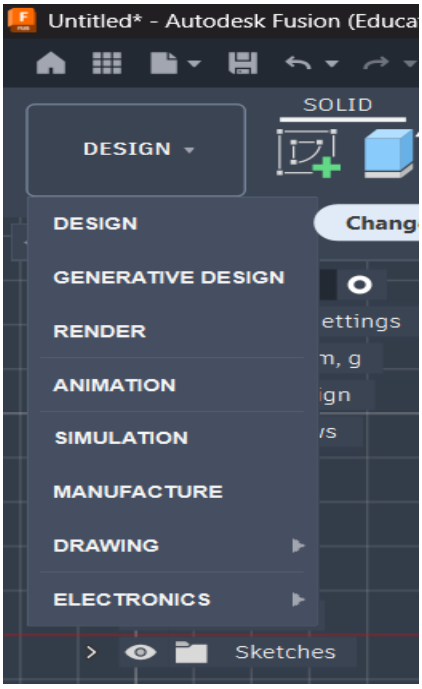
15. Now, use the orbit option to view the back of this model to create the curve.

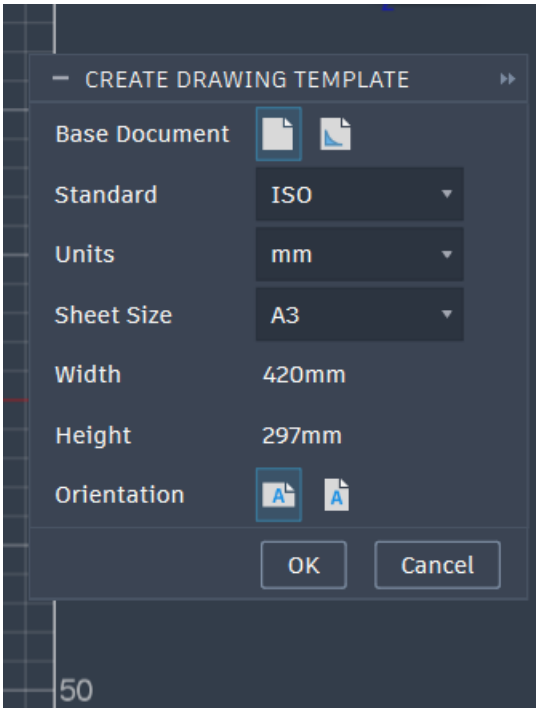
16. Use the fillet option and select the four borders of the model and give the radius value 16mm.

17. After finishing the sketch use the finish sketch option to finalize the sketch.

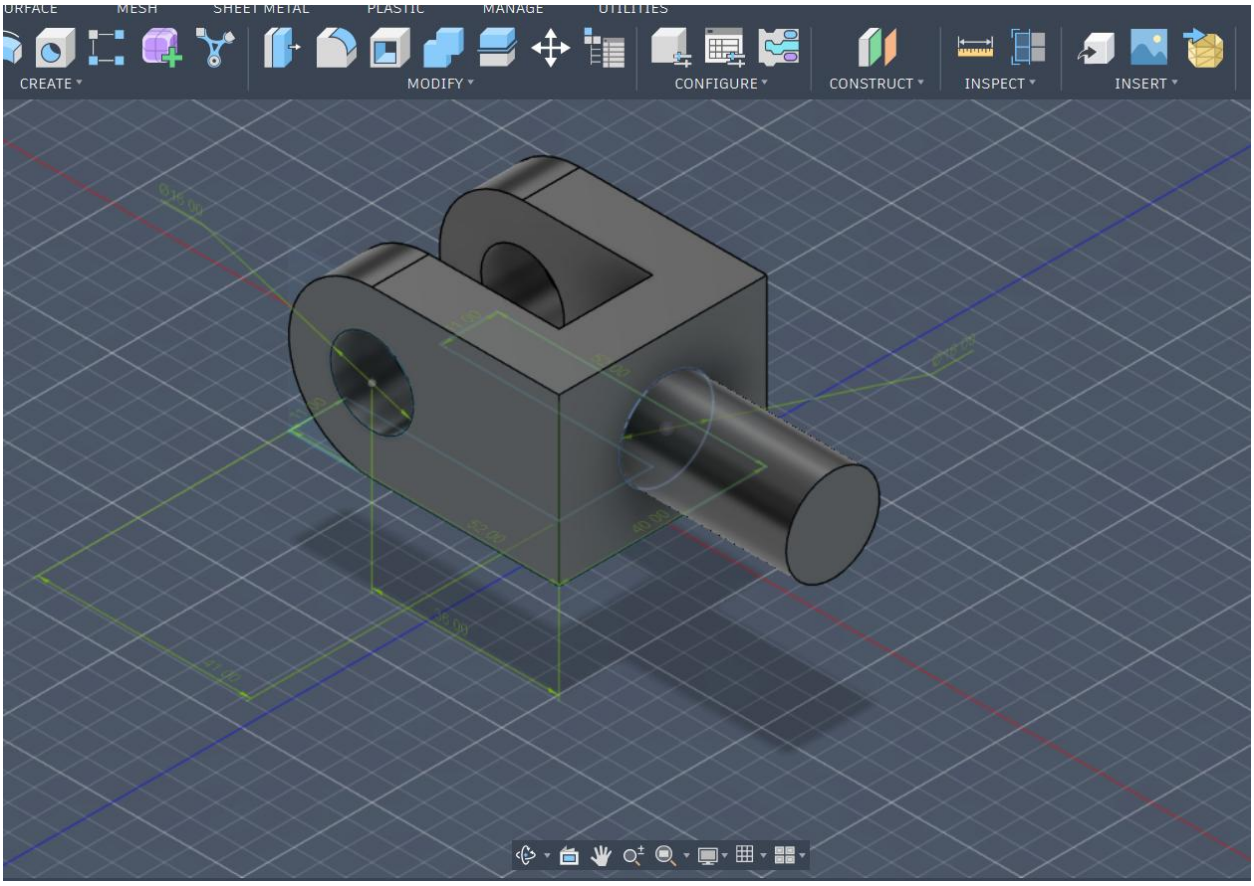


18. After that, switch to drawing mode to plot the sketch in ISO standard.

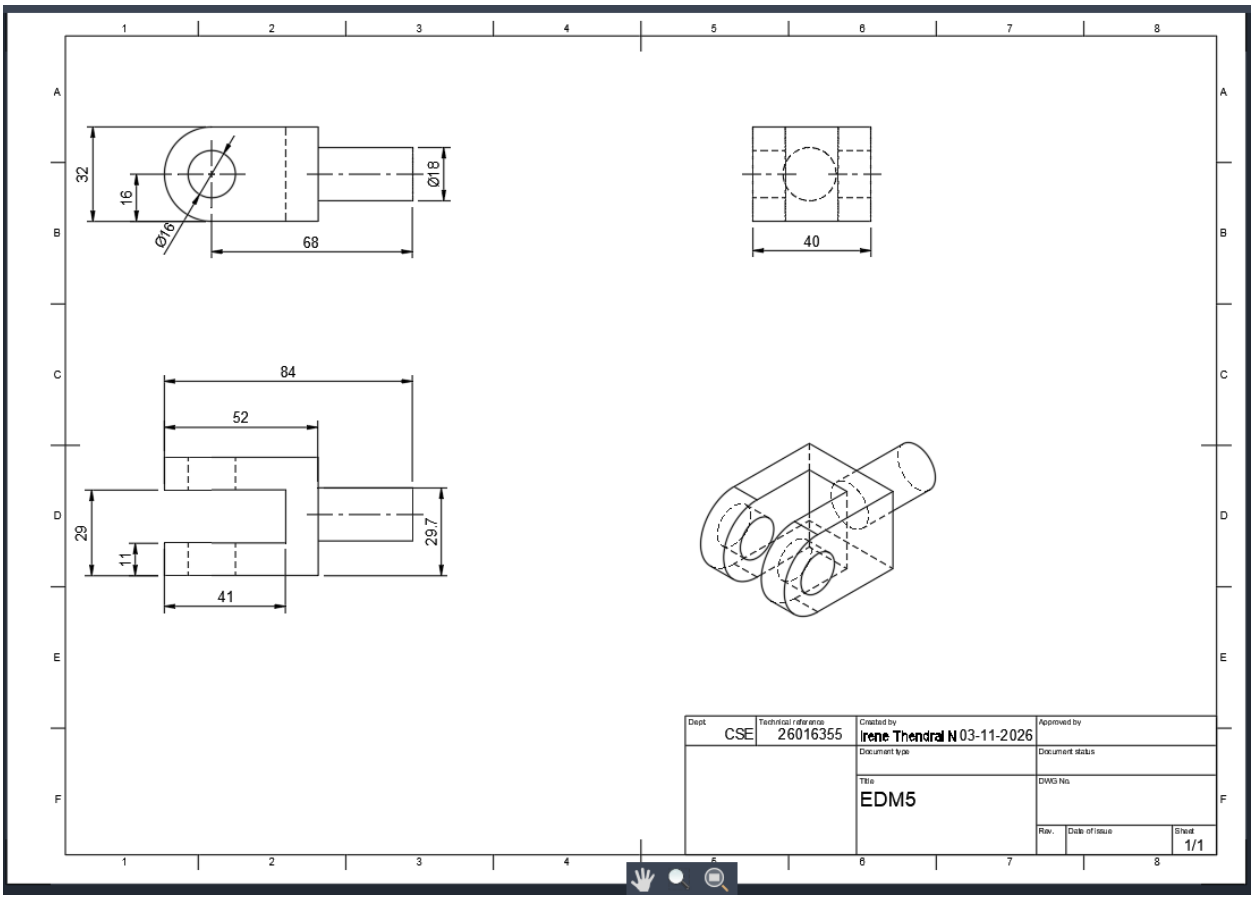



CREATE DRAWING TEMPLATE
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Standard
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Units
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Sheet Size
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Height
297mm
Orientation
OK
Cancel
50

SKETCHS:



OUTPUT:



RESULT:

Thus, the 3D model is created from the given orthographic projection.